Geophys. J. R. astr. Soc. (1977) **51**, 691–699

Spectral analysis of Quaternary palaeomagnetic data from British Columbia and its bearing on geomagnetic secular variation

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Received 1977 February 15; in original form 1976 July 16

Summary. Palaeomagnetic results obtained from a 7-m sedimentary sequence in southern British Columbia spanning approximately 9000 yr (~ 22 000 to ~ 31 000 yr BP based on radiocarbon dates) are reported and analysed. Remanence directions from 37 horizons spread throughout the section have been determined, and no evidence of the geomagnetic excursion observed at Lake Mungo, Australia (~ 31 000–28 000 yr BP), or that observed at Mono Lake, California (25 000–24 000 yr BP) is present. However, regular oscillations in the remanence vectors are observed, and maximum entropy spectral analysis, after mapping the remanence directions on to the complex plane, reveals peaks at periods of approximately 2000 and 5000 yr. The 2000-yr peak is associated with clockwise looping of the geomagnetic vector, and therefore most likely represents the time associated with one full cycle of the westward drift of the non-dipole field. The 5000-yr peak is associated with elliptical counterclockwise looping of the local geomagnetic vector and may be indicative of counterclockwise motion of the geomagnetic dipole axis.

Introduction

Since the beginning of the study of magnetization in recent sediments, interesting periodicities in the behaviour of the geomagnetic field have been noticed (Johnson, Murphy & Torreson 1948; Nagata, Hirao & Yoshikawa 1949; Mackereth 1971; Creer 1974; Creer, Gross & Lineback 1976; Opdyke *et al.* 1972). It has, in fact, become generally accepted that the study of these periodicities is critical to the understanding of the physical mechanisms underlying geomagnetic secular variation. Variations in declination and inclination have been examined in detail by many authors, but unfortunately in most cases the two directional elements have not been studied simultaneously. Where it has been possible to examine the two-dimensional oscillatory behaviour of palaeomagnetic directions, a strong tendency towards looping has often been observed (Denham 1974) as in the case of the well-known London and Paris historical records. The importance of this looping as a source of information concerning the drift of the non-dipole field with respect to the mantle was first pointed out by Runcorn (1959) and is discussed in detail by Skiles (1970). In

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essentially all cases of practical interest the motion of a non-dipole focus past any point on the Earth's surface will generate an elliptical loop in the geomagnetic vector at that point. In a normally polarized field, the sense of the loop, looking along the vector, will be clockwise if the motion is westward, and counterclockwise if the motion is eastward. Skiles (1970) points out that the existence of clockwise looping in the Japanese Narita bed data is indicative of westward drift of the non-dipole field some 50 000 yr ago. The presence of counterclockwise looping in palaeomagnetic data from Mono Lake basin in California (Denham 1974) suggests that isolated occurrence of eastward drift may also have existed in the past.

In most of the studies referenced here, the presence of either looping or periodic variation in declination or inclination could be established simply from visual inspection of the data. However, high noise levels, combined with the presence of more than one frequency in some data sets, can make visual analysis difficult. In such cases, the application of more sophisticated methods of spectral analysis is desirable. Denham (1975) outlines the particular method of spectral analysis best suited to palaeomagnetic data. The method involves mapping the data on to the complex plane about the appropriate norm, and then analysing the resulting complex time series by means of the so-called maximum entropy method of spectral analysis. The use of Burg's (1967, 1968) maximum entropy spectral estimator was suggested because of its obvious superiority over other techniques in dealing with short, relatively noisy data sets. The advantage of the complex representation of the data is that all the directional information is contained in a single time series. As a result, one can determine not only the power at a given frequency, but also the tendency towards clockwise or counterclockwise looping at that frequency.

In this paper palaeomagnetic data recently obtained from a sedimentary sequence in southern British Columbia are presented which constitute a time series of ancient geomagnetic directions spanning approximately 9000 yr. The above method of spectral analysis is applied to this data, and the resulting spectrum is discussed.

Geological and sampling details

Palaeomagnetic specimens were collected from a stream cut escarpment on the edge of Bessette Creek approximately six miles east of Lumby, BC (50° 17′ 55″ N, 118° 51′ 45″ W). The portion of the outcrop sampled consisted of dark, organic-rich, flood plain sediments containing four radiocarbon dated horizons and two tephras (Westgate & Fulton 1975). Two of the dates were obtained from detrital peat and the other two were from wood (Fig. 1). Some time may pass between the death and deposition of any biological material and thus both wood and detritus dates may be too old for their stratigraphic position. However, they cannot be too young. The wood, having retained its integrity, is less likely to have suffered a long pre-depositional history, and since both detritus dates appear too old relative to the wood dates (the upper one is even out of sequence), we feel that the sedimentation rate given by the two wood dates is likely to be the most accurate. The lowermost wood-dated horizon is at the base of the section (31 200 ± 900 yr BP). The younger horizon is 4.50 m above this and is dated at 25 300 ± 320 yr BP, giving an average deposition rate of 13 ± 2 yr/cm. It is perhaps worth pointing out that the discrepancies between the various dates are not large and this is reflected by the fact that linear regression using all four dates yields a deposition rate of 10 yr/cm.

Palaeomagnetic samples were collected at approximately 19-cm intervals starting immediately beneath the Cherryville tephra, 2.45 m above the level of the upper wood date: two specimens were taken from each of 37 horizons. Based on extrapolation of the estimated

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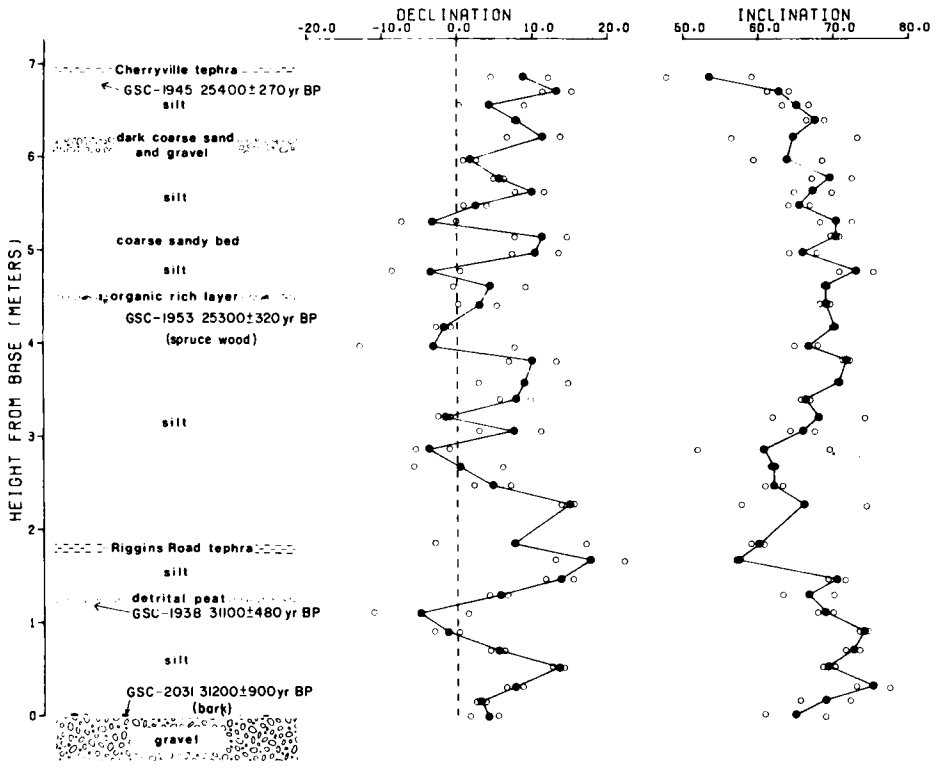


Figure 1. Magnetic directions for individual specimens (open circles) and horizon means (after AF cleaning at 100 Oe). The white regions, which make up most of the stratigraphic diagram, consist of dark flood plain silt. The two wood dates (labelled) are regarded as providing the most reliable deposition rate (see text). The other two dates shown are from detrital peat.

deposition rate, the age of the youngest specimens is 22 100 yr BP. The oldest specimens were taken at the base of the section (31 200 yr BP). With the exception of two coarse sandy beds approximately 20 cm in thickness near the top of the section, the sediment is remarkably uniform in character and there is no evidence of any major change in depositional environment during the time span sampled.

Specimens were collected by pressing small plastic boxes into the vertical face of the outcrop. The boxes were oriented using a Brunton compass, then removed and tightly capped to prevent desiccation. Two independently oriented specimens were collected from each horizon. The boxes were 2-cm square in cross-section and thus each specimen spans some two to three decades.

Measurements and results

The direction and intensity of the natural remanent magnetization of each specimen was measured using a Digico balanced fluxgate magnetometer system. Eight pilot specimens were then subjected to stepwise AF (alternating field) demagnetization in fields up to 400 Oe peak value, and a typical median destructive field was found to be about 200 Oe. The remanence directions varied by less than 10 degrees until AF treatment exceeded 150 Oe, beyond which some specimens became unstable. Consequently, 100 Oe was chosen as an appropriate field for AF cleaning of the remaining specimens. This treatment resulted in

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systematic movement away from the present field at the sampling locality, suggesting removal of small secondary components of recent origin. Prior to cleaning, intensities ranged from 2.3×10^{-4} to 5.2×10^{-6} emu/g with an average of 6.2×10^{-5} emu/g. After cleaning at 100 Oe peak field, the average intensity was 4.9×10^{-5} emu/g.

In Fig. 1, specimen directions and horizon means are illustrated in terms of declination and inclination magnetograms. The mean angular difference between pairs of specimens at each horizon is $5.9^\circ \pm 4.4^\circ$ (s.d.), this scatter most likely being due to uncertainty involved in orienting the plastic cubes, combined with minor distortion of the sediment which may occur during sampling. The standard two-tier analysis (Watson & Irving 1957) yields $k_w = 246$ (where k_w stands for the within-site precision parameter); this corresponds to a circular standard deviation (θ_{63}) of 5.2°, and thus to a circular standard deviation of the mean (α_{63}) of 3.7°. The overall mean direction for the entire sequence is $D = 005.8^\circ$, $I = +67.6^\circ$ ($N = 37$, $k = 329$, $\alpha_{95} = 1.4^\circ$) which differs significantly from the present geomagnetic vector at the sampling site ($D = 022.2^\circ$, $I = +72.6^\circ$).

The Beran & Watson (1967) statistical test for serial correlation (Epp, Tukey & Watson 1971) indicates that the probability of sequential ordering in the sequence of horizon mean directions is greater than 99 per cent, and this implies that at least some of the oscillations apparent in the magnetograms (Fig. 1) represent real temporal fluctuations in the palaeomagnetic field, the significance of which is discussed in detail below.

Spectral analysis results

Initially the declinations and inclinations were analysed separately. Equal time intervals between horizons were assumed, and both the maximum entropy and fast Fourier transform techniques were applied. In both cases, the resulting spectrum for declination was dominated by two peaks, one at a period of about 2000 yr and the other at about 900 yr; and the inclination spectrum was dominated by a single peak at approximately 5000 yr. The maximum entropy results are shown in Fig. 2. The reader should bear in mind that the plots in Figs 2 and 4 are power density plots, hence power is equivalent to the area under a peak rather than its height (Lacoss 1971). In order to obtain an acceptable spectrum using the Fourier transform technique it was necessary to apply a Hanning window to the data, consequently reducing the effective length of an already short time series. In subsequent analyses only the maximum entropy technique is used.

The average spacing between horizons is 19 cm (250 yr), but two gaps greater than 25 cm exist in the record, indicating that the assumption of uniform temporal spacing is only a fair approximation at best. In order to improve the initial spectral estimates, the data were therefore interpolated to a uniform sampling interval of 19.0 cm, prior to spectral analysis (interpolation was performed on each Cartesian coordinate using natural cubic spline functions (Greville 1967; Späth 1974)). The resulting spectra are shown in Fig. 2. The '900-yr' peak in the declination spectrum is much reduced after interpolation and may therefore be spurious. We do not feel justified in attributing a physical cause to it until resampling on a finer scale has been carried out. There remain two dominant peaks – at about 2000 yr in declination and at about 5000 yr in inclination.

As described in the introduction, the presence of open loops in secular variation data can provide important information concerning the nature of geomagnetic variations. We have therefore carried out the appropriate analysis, as described by Denham (1975). Briefly the method involves analysing both the declination and inclination information simultaneously as a complex time series. Spectra obtained from complex series are often asymmetrical in that a peak in the 'positive frequency' half of the spectrum is not the same size as its

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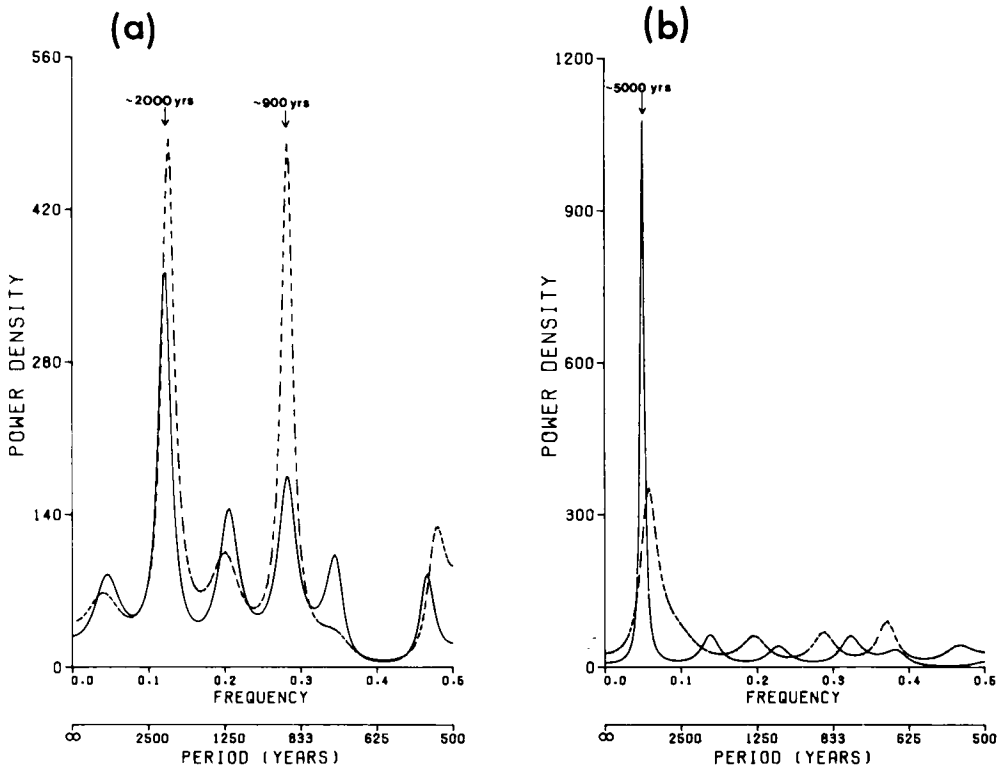


Figure 2. Maximum entropy power spectra. In each plot the dashed line represents the spectrum before interpolation, whereas the solid line is the spectrum after interpolation. The declination spectra are shown in (a), the inclination spectra in (b). The units associated with the ordinate for each plot can be thought of as (degrees)^2/unit frequency, since the amplitude of oscillation (in degrees) associated with a given peak is equal to the square root of the area under the peak.

counterpart in the 'negative frequency' half. The greater the degree of asymmetry at a given frequency, the more circular is the looping at that frequency.

In order to generate the appropriate complex series the sequence of interpolated directions was first mapped on to an angular equidistant polar projection centred on the vectorial mean of the data. Each direction was then converted to a complex number $(x + iy)$ using the transformations $x = r\cos\theta$, $y = r\sin\theta$, where r = distance (in degrees) from the centre of the projection, θ = azimuth (measured clockwise on the polar projection), and $i = \sqrt{-1}$. As the maximum entropy routine used to compute the spectra shown in Fig. 2 (Andersen 1974), was originally designed to analyse real data only, some minor modifications were necessary before analysing the complex time series.

Before describing the spectrum obtained from this analysis, it is important to point out that the ability of the maximum entropy technique to provide a good spectral estimate for a given data set, relies on the choice of the correct number of terms for the prediction error filter (PEF). Analysis of the present data using many different PEF lengths showed that filters ranging from 11 to 20 terms gave similar spectra. Experimentation with synthetic data containing known frequencies, but having a similar length and signal-to-noise ratio as the present data, indicated that the best representation of the true spectrum was obtained using PEF lengths within the above range. A more objective criterion for choosing the correct filter length has recently been put forward by Ulrych & Bishop (1975), who suggest

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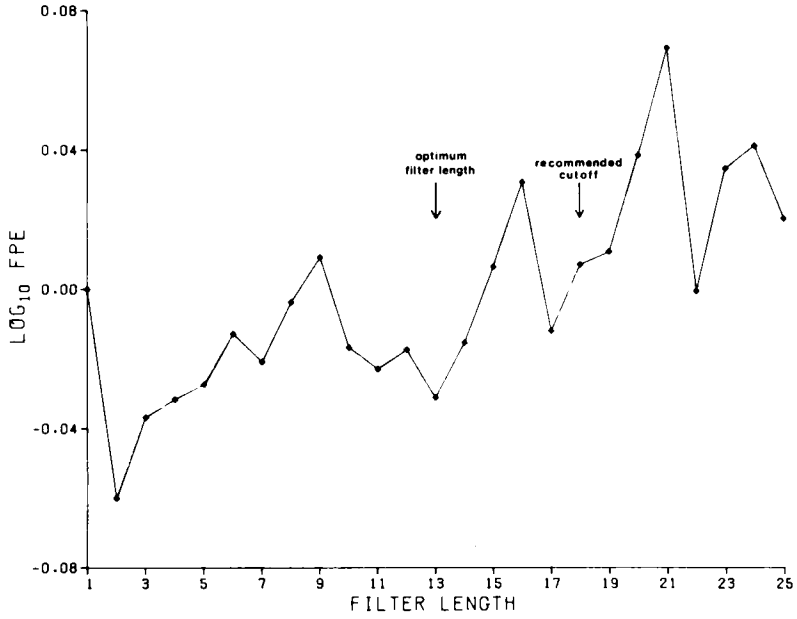


Figure 3. Plot of FPE versus prediction error filter length. Ulrych & Bishop (1975) have suggested that a cutoff of $N/2$ in the filter length be imposed when using this criterion (N = number of terms in time sequence). For the present data $N = 37$, hence FPE values for filter lengths greater than 18 are not considered.

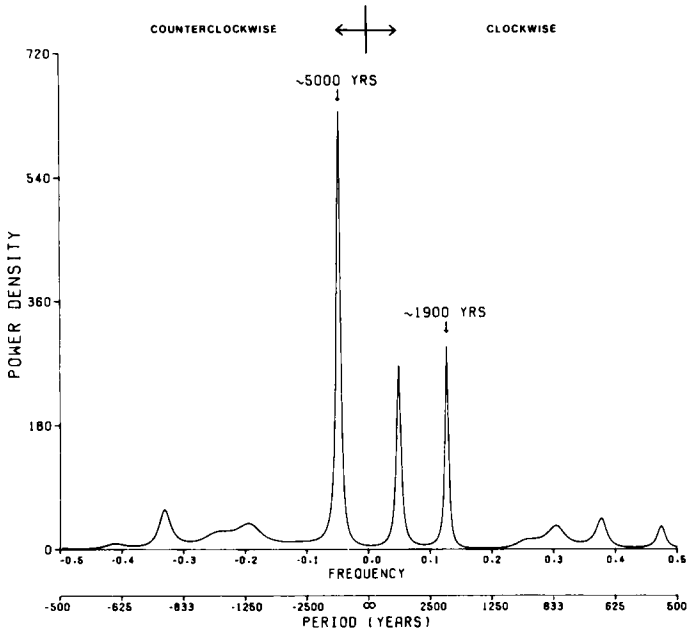


Figure 4. Spectrum of the complex time series. Units for the ordinate are the same as in Fig. 3, hence power (in degrees^2) associated with a given peak is equivalent to the area under the peak. Power in the positive frequency half of the spectrum is indicative of clockwise looping whereas counterclockwise power resides in the negative frequency half of the spectrum.

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that the optimum filter length is that which minimizes what they refer to as the 'final prediction error' (FPE). A plot of FPE versus filter length is given in Fig. 3. The position of the first minimum is surprising since a two term filter would be much too short to resolve any appreciable detail in the spectrum of the present data set. Application of this criterion to the synthetic data discussed above, indicated that in order to obtain meaningful spectra such low-order minima in the FPE had to be ignored, however the next most pronounced minimum did, in fact, seem to be a reliable indicator of the correct filter length. The second most pronounced minimum in Fig. 3 appears at 13, hence this was the PEF length used in the final analysis of the data.

The spectrum obtained in the manner outlined above is given in Fig. 4 and is clearly dominated by three major peaks. One of these strongly suggests clockwise circular looping at a period of about 1900 yr, whereas the other two are indicative of elliptical counterclockwise motion at a period of about 5000 yr. In the shorter period region of the spectrum the power present is spread over a wide range of frequencies, and neither clockwise nor counterclockwise motion appears to dominate. Further sampling on a finer scale is planned to attempt better resolution of the short-period signals.

Discussion

The magnetic stability, internal consistency, significant difference from the present local field, and high degree of serial correlation combine to strongly suggest that the data represent a reliable geomagnetic record. It is therefore interesting to note that there is no indication of large angular shifts which could be correlated with the geomagnetic excursions reported from Lake Mungo, Australia (30, 780–28, 140 yr BP: Barbetti & McElhinny 1972) or Mono Lake, California (25 000–24 000 yr BP; Denham & Cox 1971; Liddicoat & Coe 1975). This either implies that some (or all) of the sedimentation and/or magnetization dates are in error, or that depositional gaps are present, or that excursions of this kind result from the spatially restricted effects of specific non-dipole sources as envisaged by Harrison & Ramirez (1975).

The observed mean inclination is virtually identical to that of an axial dipole (+67.5°), in agreement with the secular variation model put forward by Cox (1975). At lower latitudes Cox's model predicts inclinations shallower than an axial dipole, but the inclination anomaly is expected to be close to zero at latitude 50°, where our samples were collected. Of course, zero inclination anomaly also implies that the mean virtual geomagnetic pole (VGP) will not be far-sided as anticipated by Wilson's (1970, 1971, 1972) offset dipole model. The mean of the 37 virtual geomagnetic poles (VGP's) lies at 023.1° W, 86.3° N ($K = 125$, $A_{95} = 2.1°$, $\delta = 7.2°$). This pole is statistically distinct from the spin axis, but is 'right-handed' rather than 'far-sided'. Although this tendency to 'right-handedness' is a commonly observed feature, its physical significance is still not clear (Wilson 1972). The mean VGP is also associated with rather small angular uncertainty ($\delta = 7.2°$), inviting comparison with the Pacific secular variation anomaly (Doell & Cox 1972) to which our sampling locality is peripheral. However, our data span only 9000 yr and since the time required to adequately sample secular variation and thus obtain a reliable palaeomagnetic pole is still an open question (see, e.g. McElhinny & Merrill 1975) we feel it would be premature to draw any firm conclusions concerning both this comparison, and that concerning far-sided and right-handed poles.

The presence of a spectral peak in declination at 2000 yr is intriguing for two reasons. Firstly, oscillations of similar period have been previously noted by other authors (Creer *et al.* 1976; Yukutake 1962); secondly, 2000 yr is close to the time associated with one full

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cycle of the so-called westward drift of the non-dipole field using the mean drift rate calculated from historical data (Yukutake 1962; Bullard *et al.* 1950; Adam *et al.* 1964). However, the complex MEM analysis offers a more fruitful approach so we prefer to confine discussion to the results illustrated in Fig. 4.

Although the spectral peaks at 5000 yr appear sharp, their actual widths cover a range of periods conservatively estimated as extending from 4000 to 7500 yr. In other words, the period of oscillation represented by these peaks is not significantly different from the 6000 yr period discovered by Opdyke *et al.* (1972) in magnetic inclination recorded by sediments from the Aegean Sea. These authors suggest that this periodic variation may be due to a counterclockwise circular motion (12° radius) of the geomagnetic dipole axis around the spin axis. Our data definitely imply counterclockwise motion, but of a highly elliptical nature and smaller amplitude (~ 6° semi-major axis).

The spectral peak at 1900 yr covers a range of periods from about 2100 yr to about 1800 yr. The strong asymmetry indicates that the motion involved is circular and clockwise. It presumably represents the period associated with one full cycle of the westward drift of the non-dipole field. If this interpretation is correct, it implies that at least some features of the drifting non-dipole field are relatively stable for 2000 yr or more, and that the phenomenon of westward drift was in existence some 30 000 yr ago.

Acknowledgments

We sincerely thank Dr R. J. Fulton and Dr J. A. Westgate whose geological advice and guidance in the field made this study possible. We are also very grateful to Dr J. W. Olson and Dr R. G. Currie for advice regarding the application of spectral analysis. This work was financed by a grant from the National Resarch Council of Canada.

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